

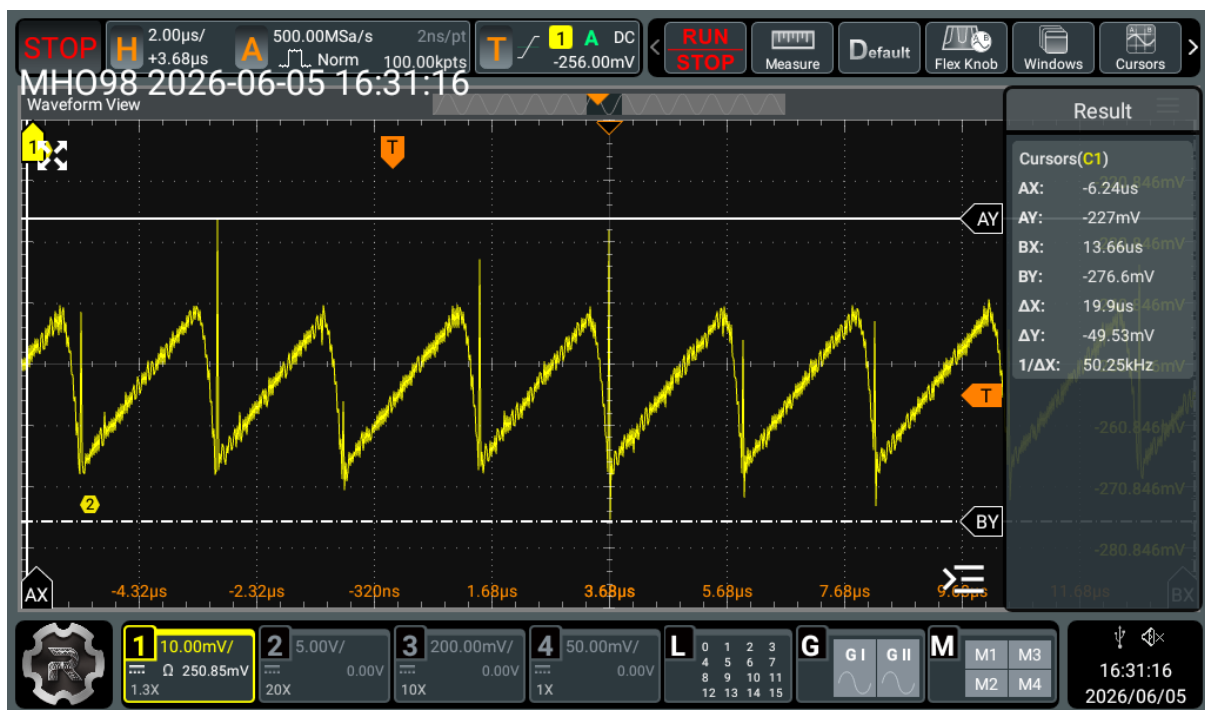
Here are some measurements that I was just doing on a board.

During EMC testing, we noticed some noise around 175MHz, that was still within the limits, but wanted to address it anyway. With near field probing we found a DC-DC converter that was making this noise on it's input. This was a 5V rail, so I measured it first with a regular probe, on an 8 bit scope, just to see that the radiated noise from the inductor, quantization noise probing etc, making this measurement impossible.

<https://www.analog.com/en/resources/technical-articles/input-and-output-noise-in-buck-converters-explained.html>

The noise appears when the FET is switching, as explained on Figure 9 of the paper above. For the analog devices test, the ringing seems to appear on the high side FET, for me it was the low side FET.

Here are the pictures of the switching noise on my board:

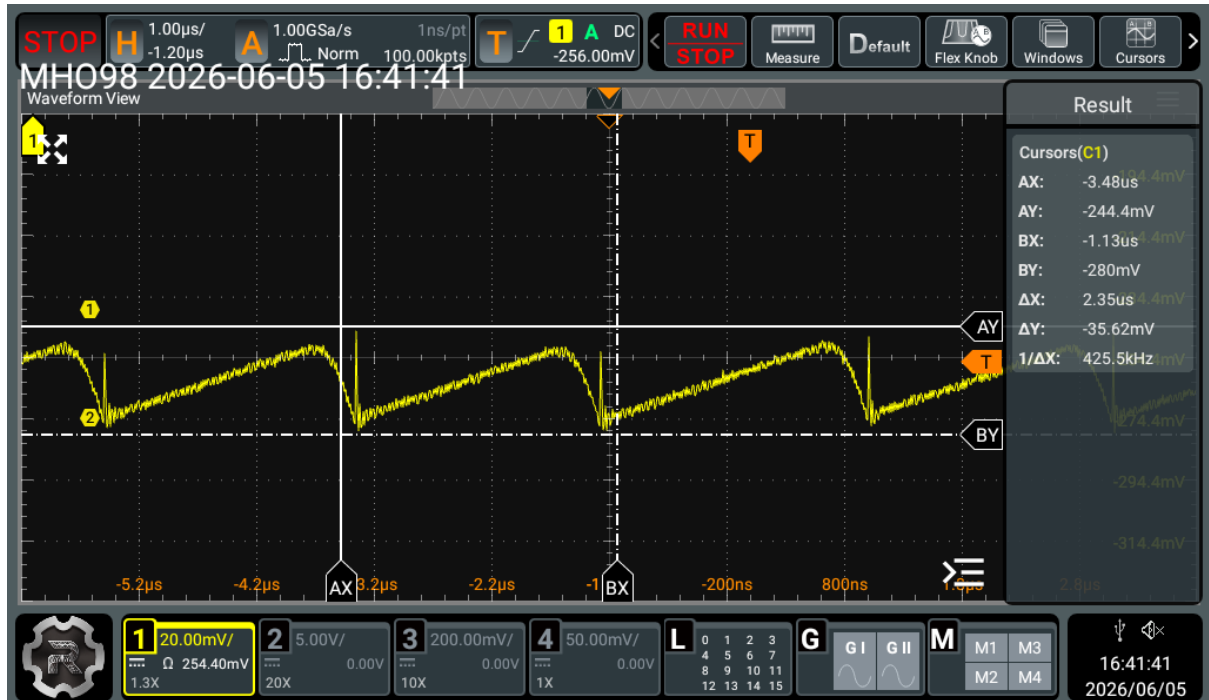


Switching ripple before modification



High frequency ringing before modification.

We've added an extra capacitor, 10nF, with the SRF at around 150MHz on the input. This reduced the amplitude of the noise by about 40%. Here are the after images:



Switching ripple after modification



High frequency ringing after modification.

We also tested the circuit before and after in a TEM cell and the radiated noise got reduced by 20dB, so now this is way below the limits required by the standards for compliance.

The PRP1 made these tests possible to be performed in the time domain.